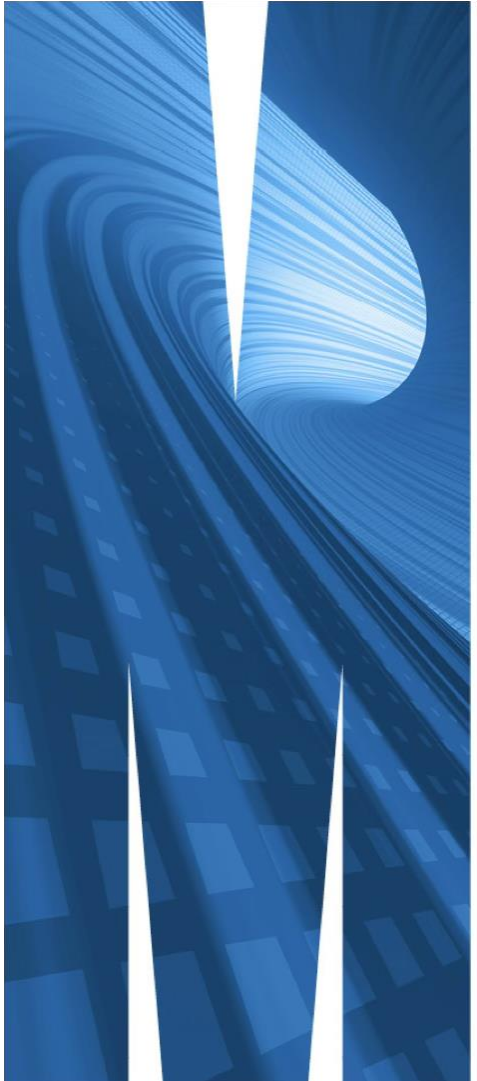
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FIT3179 Data Visualisation

Week 6: Network and Tree Idioms
Data format - Activity

Network Data

May already be in network format:

- Adjacency matrix
- Node list + edge list (JSON, CSV)
- Specific graph file formats: Graphml, GML, GEXF

Or may require wrangling to turn into a network:

- Tabular data (CSV, Excel)

Adjacency Matrix

	A	B	C	D	E	F
A	0	1	1	0	0	1
B	1	0	0	1	0	1
C	1	0	0	1	1	0
D	0	1	1	0	0	1
E	0	0	1	0	0	0
F	1	1	0	1	0	0

Edge list

Source	Target
A	B
A	C
A	F
B	D
B	F
C	D
C	E
C	F
D	F

Graphml

```
<graphml xmlns="http://graphml.graphdrawing.org/xmlns">
<graph id="MyConnectome" edgedefault="undirected">
<node id="1">
 <data key="dn_label">VisualArea1_RH</data>
 <data key="dn_node_position">(23.13,23.34,23.76)</data>
 <data key="dn_uri">http://www.connectome.ch/wiki/V1_(Homo sapiens)</data>
 <data key="dn_correspondence_id">1</data>
</node>
...
<edge id="e5_14" source="5" target="14">
 <data key="de_number_of_fibers">10</data>
 <data key="de_average_length_mm">22.6466</data>
</edge>
...
</graph></graphml>
```

Network data within tabular data

Most of the time you are working with tabular data, however there may be relationships present in this data

Tabular data to network data: Example 1

StudentName	CourseName
John Doe	Introduction to Computer Science
John Doe	Calculus I
Jane Smith	Introduction to Computer Science
Emily Jones	Calculus I
Emily Jones	Data Structures
Michael Brown	Introduction to Computer Science
Michael Brown	Physics I

What are the nodes?

In this case, the unique students and course names:

John Doe
Jane Smith
Emily Jones
Michael Brown

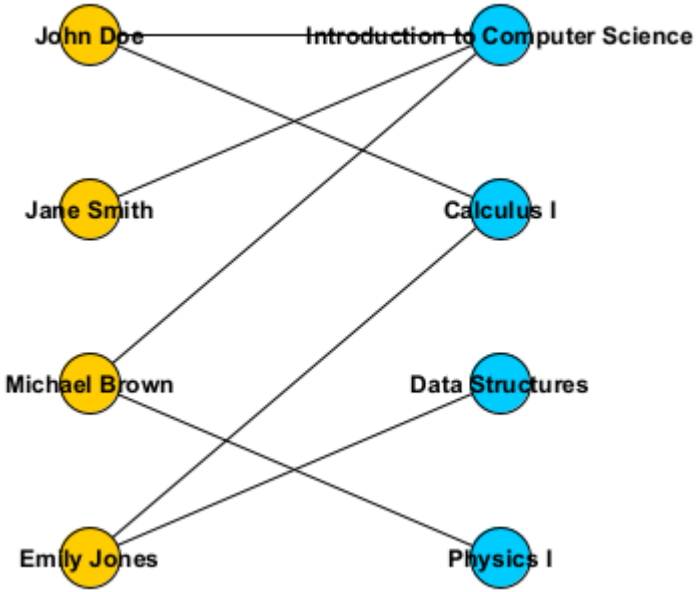
Introduction to Computer Science
Calculus I
Data Structures
Physics I

What are the edges (relationships)?

Students take particular courses (or conversely, a course is taken by particular students).

Each row in the table is thus an edge.

It’s common for nodes in networks to belong to different groups or clusters



Identifying network data from tabular data

- Requires two (typically categorical) columns, which have some kind of relationship
- Collectively, the unique values from both columns are the nodes in the network
- Each row then gives an edge where the value in column 1 is the source node and the value in column 2 is the target node
- Other columns may give additional information which can be mapped to channels. E.g. edge weight, node size

Activities

Activity 1: Tabular data to network data: Example 2 - which pairs of columns contain relational data?

Employee ID	Employee Name	Department	ProjectID	Collaboration With	Collaboration Frequency	Performance Rating
1	Alice Johnson	Engineering	P001	Bob Smith	3	4.5
1	Alice Johnson	Engineering	P002	David Lee	2	4.5
2	Bob Smith	Engineering	P001	Alice Johnson	3	4.7
2	Bob Smith	Engineering	P003	Eve Davis	1	4.7
3	David Lee	Marketing	P002	Alice Johnson	2	4.2
3	David Lee	Marketing	P003	Eve Davis	3	4.2
4	Eve Davis	Marketing	P003	Bob Smith	1	4.8
4	Eve Davis	Marketing	P003	Alice Johnson	3	4.8

Activity 2: Match the type of network to the most suitable idiom

Network Type

- Social network
  E.g. Facebook
- Hierarchical network
  E.g. World Population, grouped by continent, country, city
- Flow Network
  E.g. Supply Chain

Idiom

- Sankey Diagram
- Node-link diagram
- Treemap

Activity 3: Try it yourself

- Network files available on Moodle, in different formats. Use the karate_edges.csv and karate_nodes.csv files.

- Go to https://flourish.studio/
  - Free account - you can use your student account to quickly sign up with Google
  - Browse the different network visualisation idioms - these come with example datasets loaded which are suitable for each idiom. Look at the kind of data in the flourish defaults.
  - Try uploading the karate network to create different types of network visualisations. Which is the most suitable for this data?

Activity 4: Channels for Node-Link Diagrams

Imagine a network of social media influencers.

Two influencers are connected by the number of shared followers they have.

The total number of followers for each influencer is given, as well as the type of content they create.

- Download the influencers_edges.csv and influencers_nodes.csv files from moodle
- Go to https://flourish.studio/
- Create a visualisation of the network.
- Which channels can be used to encode all of the data described above?

InfluencerName	Followers	Content_Type
Alex_Fit	150000	Fitness
Sam_Cook	180000	Cooking
Jess_Music	220000	Music
Lee_Gaming	170000	Gaming
Mia_Life	140000	Lifestyle
Ryan_Travel	160000	Travel
Eva_Style	200000	Lifestyle
Taylor_Fit	145000	Fitness
Chris_Travel	165000	Travel
Lily_Cook	175000	Cooking

Source	Target	Shared_Followers
Alex_Fit	Sam_Cook	18795
Alex_Fit	Jess_Music	3860
Sam_Cook	Lee_Gaming	41158
Sam_Cook	Mia_Life	47732
Jess_Music	Ryan_Travel	14284
Jess_Music	Eva_Style	9265
Lee_Gaming	Alex_Fit	19850
Mia_Life	Jess_Music	40194
Ryan_Travel	Eva_Style	24962
Eva_Style	Sam_Cook	47131
Alex_Fit	Taylor_Fit	19023
Taylor_Fit	Jess_Music	44090
Sam_Cook	Lily_Cook	4685
Lily_Cook	Eva_Style	3769
Ryan_Travel	Chris_Travel	5433
Chris_Travel	Mia_Life	8311
Eva_Style	Mia_Life	40819
Jess_Music	Lee_Gaming	42188